

CONSTANT SPEED STORY: 3-STAGE DUAL-GRAPH EDITION

UNIT 2: KINEMATICS • 3-STAGE JOURNEY MODELING • DAY 15 PERFORMANCE TASK

Name: _____ Period: _____ Date: _____

SCORE: _____ / 10 pts

Mission: Invent an original character/vehicle journey across **3 distinct motion sections** (like the Dual-Graph Studio). Illustrate the scene, author your narrative, and calculate the velocities in the table. On the back, calibrate your axis scales and prove your math with both graphs!

HS-PS2-1 • PERFORMANCE TASK

PART 1: 3-STAGE JOURNEY ILLUSTRATION

Sketch the object, path of motion, origin ($x = 0\text{m}$), pause/turning point, and final point

[Draw your character, vehicle, drone, rover, or athlete traveling across 3 distinct stages]

Label: Starting Reference Point (x_0) • Section 1 Path ($\rightarrow$ or $\leftarrow$) • Section 2 Stop/Action Point • Section 3 Final Destination

PART 2: ORIGINAL 3-SECTION MOTION NARRATIVE

Describe what occurs in each phase of your object's motion:

Section 1 (0s to t_1) Initial Motion

Starts at $x = \text{_____m}$, travels to $x = \text{_____m}$:

Section 2 (t_1 to t_2) Mid-Course Action

Action (e.g. stopped/waiting/steady):

Section 3 (t_2 to t_{final}) Final Stretch

Travels from $x = \text{_____m}$ to $x = \text{_____m}$:

PART 3: 3-STAGE MOTION BREAKDOWN & G.U.E.S.S. MATHEMATICAL PROOF

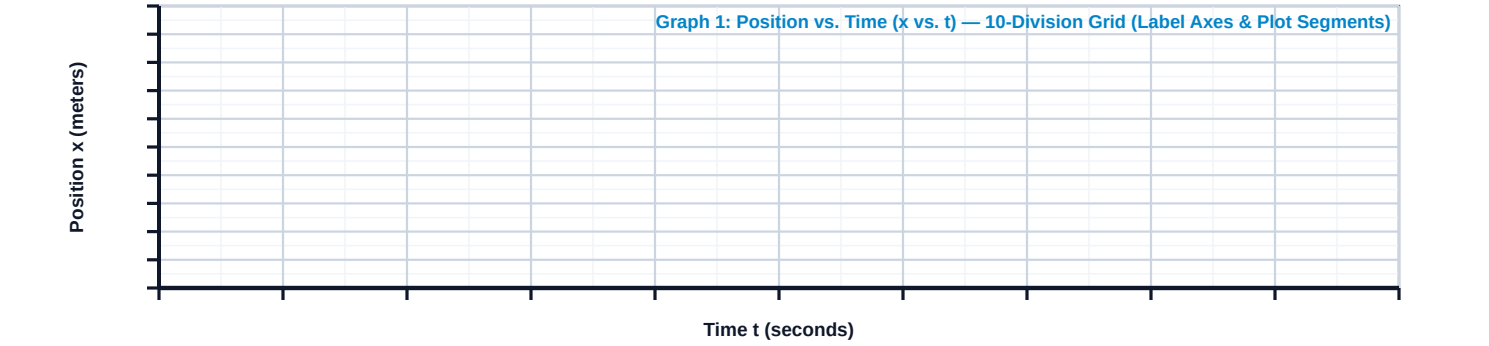
Compute displacement (Δx) and velocity ($v = \Delta x / \Delta t$) for each section

Measurement / Step	Section 1	Section 2	Section 3	Whole Trip (Total)
Time Interval (Δt) $\Delta t = t_{\text{final}} - t_{\text{initial}}$	From _____s to _____s $\Delta t_1 = \text{_____ s}$	From _____s to _____s $\Delta t_2 = \text{_____ s}$	From _____s to _____s $\Delta t_3 = \text{_____ s}$	$\Delta t_{\text{total}} = \text{_____ s}$
Position Coordinates Initial (x_i) $\rightarrow$ Final (x_f)	$x_i = \text{_____ m}$ $x_f = \text{_____ m}$	$x_i = \text{_____ m}$ $x_f = \text{_____ m}$	$x_i = \text{_____ m}$ $x_f = \text{_____ m}$	Start $x_0 = \text{_____ m}$ Final $x_{\text{end}} = \text{_____ m}$
Displacement (Δx) $\Delta x = x_f - x_i$ (include + / -)	$\Delta x_1 = \text{_____ m}$	$\Delta x_2 = \text{_____ m}$	$\Delta x_3 = \text{_____ m}$	Net $\Delta x = \text{_____ m}$
Direction of Motion	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	☐ Forward (+) ☐ Stopped (0) ☐ Backward (-)	Total Distance: $d_{\text{total}} = \text{_____ m}$
G.U.E.S.S. Velocity Formula $v = \Delta x / \Delta t$ (Substitute & Solve)	$v_1 = \text{_____m} / \text{_____s}$ $v_1 = \text{_____ m/s}$	$v_2 = \text{_____m} / \text{_____s}$ $v_2 = \text{_____ m/s}$	$v_3 = \text{_____m} / \text{_____s}$ $v_3 = \text{_____ m/s}$	Average Speed: $v_{\text{avg}} = d_{\text{tot}} / \Delta t_{\text{tot}}$ $= \text{_____ m/s}$

DUAL-GRAPH MOTION MODELING • 3-STAGE PROOFS
SLOPES EQUAL VELOCITIES • AREAS UNDER THE CURVE EQUAL DISPLACEMENTS

Name: _____ Period: _____

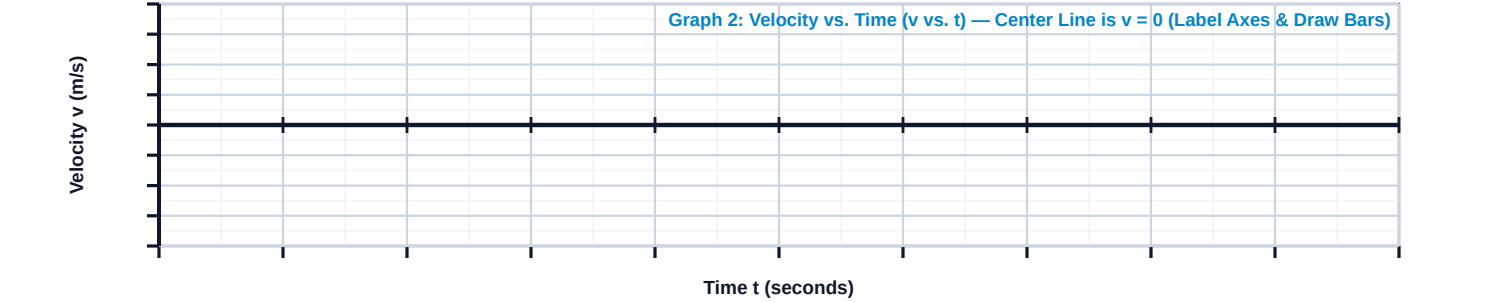
Axis Scale Calibration: Choose your scale to fit your story • Write your numbers along the axis tick marks before plotting: (Aligned 10-Division Grid)
Time Axis (t): 1 block = _____ s Position Axis (x): 1 block = _____ m Velocity Axis (v): 1 block = _____ m/s



Section 1 Slope Proof: $m_1 = \Delta x_1 / \Delta t_1 = (\text{_____m}) / (\text{_____s})$
Slope₁ = _____ m/s (Matches v_1 ?)

Section 2 Slope Proof: $m_2 = \Delta x_2 / \Delta t_2 = (\text{_____m}) / (\text{_____s})$
Slope₂ = _____ m/s (Flat line = 0?)

Section 3 Slope Proof: $m_3 = \Delta x_3 / \Delta t_3 = (\text{_____m}) / (\text{_____s})$
Slope₃ = _____ m/s (Matches v_3 ?)



Section 1 Area (Δx_1):
Area₁ = Height $\times$ Width = (v_1)(Δt_1)
= (_____ m/s) $\times$ (_____ s) = _____ m

Section 2 Area (Δx_2):
Area₂ = Height $\times$ Width = (v_2)(Δt_2)
= (0 m/s) $\times$ (_____ s) = 0 m

Section 3 Area (Δx_3):
Area₃ = Height $\times$ Width = (v_3)(Δt_3)
= (_____ m/s) $\times$ (_____ s) = _____ m

PART 4: DUAL-GRAPH SYNTHESIS & NET DISPLACEMENT VERIFICATION

1. Total Net Displacement Proof from Shaded Areas:
Sum of all 3 areas: $\Delta x_{\text{total}} = \text{Area}_1 + \text{Area}_2 + \text{Area}_3$
= (_____ m) + (0 m) + (_____ m) = _____ meters
Does this match your position change on Graph 1 ($x_{\text{final}} - x_{\text{initial}}$)? ☐ YES ☐ NO

2. Physical Meaning of Negative Slope / Area:
When a line on Graph 1 slopes downward (negative slope), what direction is the object moving? _____
Where does its horizontal line appear on Graph 2? _____

3. Unit Cancellation Proof: In Graph 2, show mathematically how multiplying the height unit (m/s) by the width unit (s) cancels to prove that area MUST be distance: (m / s) $\times$ (s) = _____

CORE CONCEPT CHECK